Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Onboard Carbon Capture (OCC) as a decarbonization option for shipping			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility (), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine (√), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical・Materials		Chemicals	
	Medium Category	Small Category	Medium Category	Small Category
	Shipbuilding/ Marine Systems	Other Shipbuilding/Marine System Technologies	Air / Waste	Waste Treatment and Recycling Technologies
Concept of Proposed Technology	Enabling the implementation of onboard carbon capture (OCC) for vessels with internal combustion engines running on HFO, MGO, LNG, methanol etc.			
Rationale for Joint R&D Project	Factors like the type of vessel, sailing profile, engine and fuel type determine the optimal onboard carbon capture configuration. Matching technology and application requires collaboration between vessel operator and research institute			
Key Words	1. Vessel energy system design 2. Infrastructure development 3. Digital twin 4. Emission profiles 5. Pilot project			

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (5-6)	End of the Project	Level (7)
R&D Objectives	Design optimal onboard carbon capture configurations for dedicated use-cases. Example issues (see next slides) to resolve: NO2 in exhaust gas, avoid/reduce aerosol-based emissions. Other aspects that can be assessed: methane slip reduction, heat integration. Having a digital twin of the vessels energy system allows for OCC optimization			
R&D Target Specifications	Minimize fuel penalty and CAPEX of OCC systems			
R&D Plan	Digital Twin of energy system → Design of optimal OCC solution → piloting OCC onboard at TRL6-7			

4. R&R (Roles and Responsibilities)

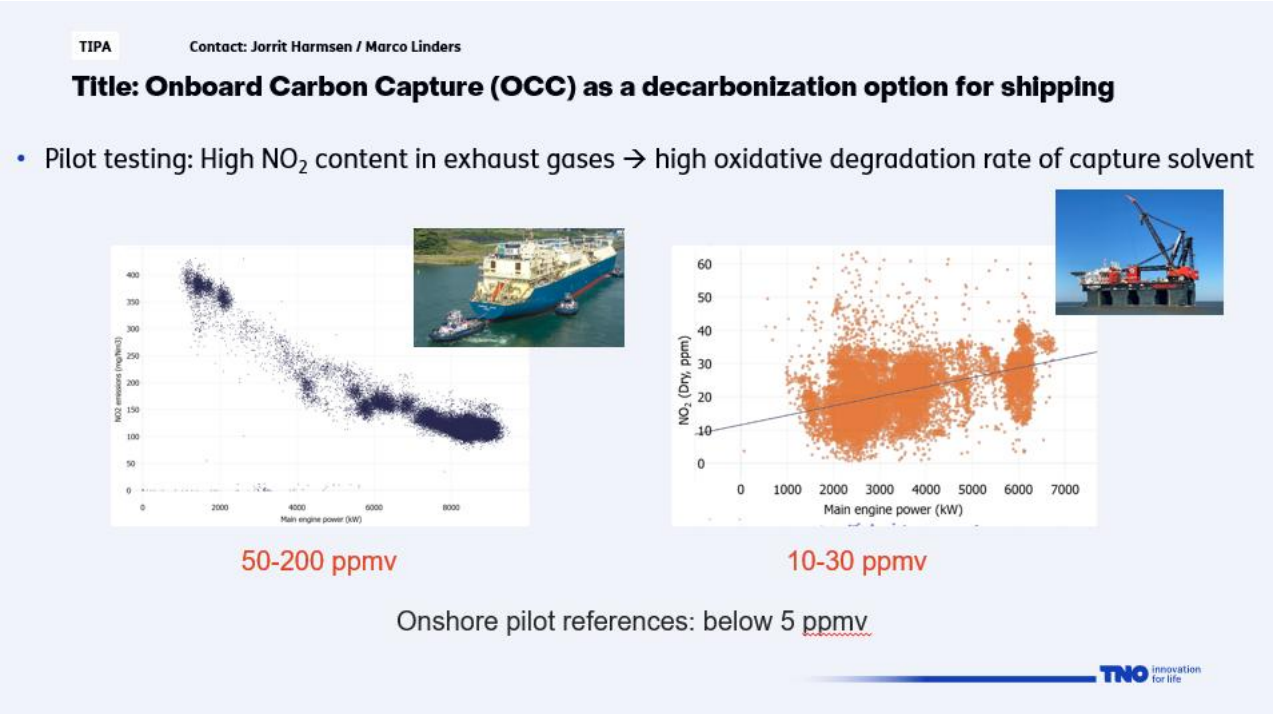
Expected R&R of Research Institute	Digital Twin, data analysis, emissions measurements, full-scale OCC design, pilot design
Expected R&R of Korean SME Partner	Provide data, provide vessel for piloting, operate pilot, support data analysis

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Validated design of an OCC system capable of meeting IMO's target with minimized fuel penalty and CAPEX

6. References

1) J. Ros, E. Skylogianni, V. Doedee, J. van den Akker, A. Vredeveldt, M.J.G. Linders, E. Goetheer, and J. Monteiro, Advancements in ship-based carbon capture technology on board of LNG-fuelled ships. International Journal of Greenhouse Gas Control 114, 2022. (DOI: 10.1016/j.ijggc.2021.103575)
2) Project REMARCCABLE – a case study for onboard carbon capture on the MR tanker Stena Impero, M. Traver, J. Ros, J. Monteiro et al.; The Oil and Gas Climate Initiative, 2024. OGCI | Project REMARCCABLE | Report
3) EverLoNG - Demonstration of ship-based carbon capture on LNG fuelled ships, Final report, J. Monteiro, R. Skagestad, J. van den Akker, P. Zapp, E. Vroegrijk, R. Stevenson, M.J.G. Linders, 2025. Results | EverLoNG

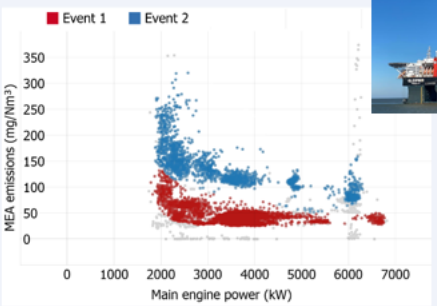


TIPA Contact: Jorrit Harmsen / Marco Linders

Title: Onboard Carbon Capture (OCC) as a decarbonization option for shipping

- Pilot testing: Potential for aerosol-based emissions → Particle measurements onboard & integrate countermeasure in OCC design

Volatile emissions dominate
Average: 2.1 mg/Nm³
Standard deviation: 14.8 mg/Nm³



Aerosol emissions dominate

TNO innovation for life

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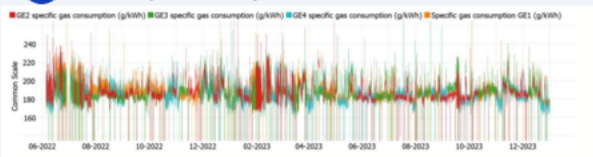
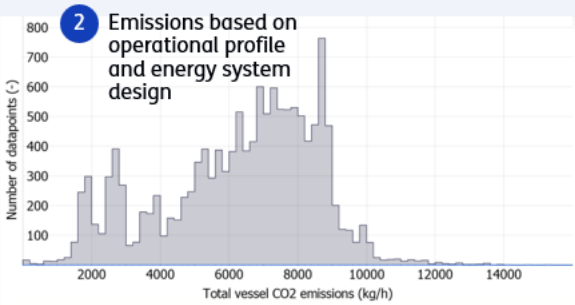
Title: Onboard Carbon Capture (OCC) as a decarbonization option for shipping

- Digital twin of vessels energy systems

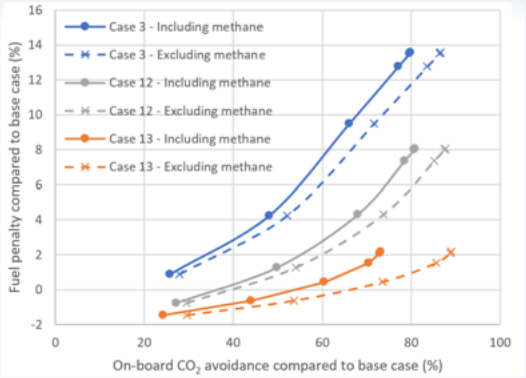
1 Vessel operational profile



2 Emissions based on operational profile and energy system design



3 Fuel penalty and CO_2 avoidance for different design options



4 Optimized OCC design for specific use-case

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